

Causal Value Models for Adaptive Prompting

Audited theory, corrected experiments, and remaining validation

DTR-MultiRoundLLM research development package

20 September 2026

Contents

Continuation report: theory and experiment audit	2
Causal value models for adaptive prompting: estimands, identification, estimation, and decisions	4
Stopping, prefix replay, and candidate selection: a scoped theory addendum	18
Corrected E0 simulation study, 2026-09-20	25
Corrected fixed-bank selection pilot, 2026-09-20	27

Continuation report: theory and experiment audit

20 September 2026. The work continued with a fresh repository audit, corrections, proofs and additional CPU experiments. The project remains **not submission-ready**. Its supported goal is history-conditional next-prompt value and policy evaluation; an individual conversation’s unobserved causal effect is not identified.

What was completed

- Added ten scoped theory results/propositions with proofs for full-rollout stopping, changed-policy weights, public-information optimal stopping, regret, censoring, candidate selection, finite-class evaluation, ESS and hidden state. Ten finite-law identity/counterexample checks pass.
- Repaired the E0 first-action weight, continuation-target comparison, template-mixture truth, false optimality label and replication accounting. Ran all **17 cells $\times$ 80 = 1,360 corrected replications**. A separate internal auditor ran 72 fitted-Q replications. Original 400-replication fixed-nuisance reference results remain a separate study.
- Independently reanalysed **4,488 existing real-model episodes / 561 tasks**. Retained first candidates from conversations that later switched receiver; evaluated the fixed-receiver repair regime with recorded routing weights. The earlier blanket “K1 fired on both models at lower cost” conclusion is withdrawn.
- Fit two public-feature selectors on 336 roots and evaluated on 225 roots, using four initial candidates per receiver/task. The 7B logistic fit improved recorded full-test success by **2.85 percentage points**, with an exploratory task interval **[1.19, 4.52]**. All four model/learner comparisons are reported; this reused corpus is not confirmatory.
- Rechecked ten load-bearing literature records. All identifiers were real, but several descriptions and novelty/possibility claims were wrong. CausalCollab and language-action DTR learning are direct predecessors. No “first causal conversation” or “first randomized LLM system” claim is made.
- **51 tests pass**. Newly added checks concern actual estimator, cohort, propensity, stopping and inference failures. All new work used local CPU, **zero new model calls, zero GPU hours and \$0 external spend**.

Findings that change the interpretation

The original routing comparison selected easier candidates by discarding conversations that later switched receivers. It also mixed different task denominators. After correction, resampling remains strong for 3B; the 7B result depends on sample count and cost. These are exploratory sibling routing-policy results, not randomized user-prompt effects.

The old E0 conditional-regression and IPW estimates did not always target the same continuation. Its state-only “optimum” was a heuristic, and its template-heterogeneity coverage used the wrong truth. In the repaired grid, weak overlap still yields poor Wald coverage (.8125 at 80 replications). Template variation alone does not break identification of an explicitly fixed template mixture. Fitted propensity robustness and precise final coverage calibration remain untested.

Full-rollout stopping evaluation has weight one only when the target uses the same continuation mechanism before stopping and the stop outcome is the recorded prefix utility. Complete prefix grades do not identify effects of untaken prompts. Naturally stopped routing logs cannot reveal missing suffixes. An ESS lower bound of 2.75 is not the actual effective sample size; a conservative certificate’s failure is not a general impossibility theorem.

The new positive selection result concerns choosing among four already-generated answers. It supplies a reason to test that specific selector afresh, but does not establish multi-round feedback gains, a causal critic, prompt optimization, or an advantage over a cheaper early-stopping baseline. The exact feature schema and training procedure are frozen in the run manifest; the discovered unused assertion-category feature will be fixed prospectively, without quietly changing this result.

A further experiment-workstream update (0ee7666) arrived during this audit. Its new execution-agreement/allocation studies retain the same future-switch exclusions and add probe-usability selection; their proposed stop decision therefore needs the [latest audit](#). The oracle opportunity bound is valid for the same bank, but does not make only 33 roots the independent sample size or prove a research-wide impossibility. Its near-one-call allocation target also differs from the new four-call selector pilot.

Remaining work, in order

1. **Accept the corrections and freeze one intervention design.** Reconcile candidate-first randomized selection with class-first generation; these have different laws and weights. Adopt the stopping assumptions and Q11 scope in the new

theory reconciliation.

2. **Finish simulation precision and fair baselines.** Prespecify Monte Carlo precision; expand the corrected grid as needed and include correctly history-adjusted sequential regression for the same target. Fit propensities only if an observational-estimation claim is retained.
3. **Validate the selection lead on new tasks.** Audit family structure and visible/hidden separation, freeze the logistic/visible/BoN comparisons, and measure total generation/checking/selector cost. Compare four-candidate banks first; prespecify a separate matched-budget adaptive comparison.
4. **Collect actual prompt-intervention evidence.** With one pinned receiver, freeze a small exact candidate slate, randomize supported next prompts at restorable histories, and name the continuation policy. Record failures and assignment probabilities; use independent roots for evaluation. This is the missing experiment for the original personalized-prompt question.
5. **Only then train and evaluate broader regimes/generators.** Assess conditional calibration and supported policy improvement. A changed generator needs fresh data or a valid full generator likelihood ratio. Do not use local branches as a substitute for whole-policy target-history validation.

No paid service or additional model inference was launched: the informative next action was to repair and independently audit existing evidence. The prospective collection remains genuinely outstanding; it is not being presented as completed by this CPU continuation.

Evidence map

Artifact	Role
Original theory and new addendum	General prompt estimand and scoped stopping/selection proofs
Theory reconciliation	Fourteen findings, replacements and Q11 resolution
Corrected E0	Fitted-Q simulation results, MCSE and reproducibility
Independent experiment audit	Raw-log integrity, corrected routing values/costs and E0 audit
Selection pilot	All four frozen-split fits and limitations
Literature recheck	Primary-source links and corrected interpretation
Execution handoff	Ordered work packages and GitHub issue links

This report and the updated PR are reviewable outputs, not evidence that another agent has accepted the handoff or that the manuscript is submission-ready.

Causal value models for adaptive prompting: estimands, identification, estimation, and decisions

Version: 2026-09-19. **Status:** results proved below under the stated assumptions; initial mathematical audit completed, independent review pending. This is an initial research theory package, not a claim of new foundational DTR theory or completed empirical validation. Companion audit: [theory_proof_audit.md](#).

1. Scientific question and contribution boundary

For a fixed receiver LLM, tool environment, task distribution, and evaluator, what would happen if the next feedback prompt were changed, and which supported adaptive prompting policy improves final task performance after accounting for cost? The intended deployed system observes a request and the receiver’s answer, generates possible feedback, scores their downstream values, chooses feedback or STOP, and repeats. We call the score a **causal prompt value model** only when the intervention, continuation policy, outcome, and identification assumptions are explicit.

The object is a history-conditional expected counterfactual outcome or contrast, not an individual’s unknowable realized treatment effect. A neural value predictor and a generative actor do not themselves establish causality. Randomization or defensible sequential exchangeability establishes identification; function approximation estimates an identified functional; independent evaluation establishes whether decisions improve.

The core DTR, g-computation, importance weighting, augmentation, and policy improvement arguments are established statistical ideas. This package specializes and connects them to a two-stage language intervention: **generate a candidate slate, then randomize a selected candidate**. Potential research contributions are the support-aware language intervention design, history-conditional learning and calibration, candidate-generation versus selection error decomposition, and empirical evidence that the resulting decisions improve over adequately adjusted alternatives. Novelty requires a separate literature audit. Merely using a Transformer, calling a critic causal, or applying DTR to prompts is insufficient.

2. Timeline, unit, and exact intervention

2.1 Unit and time origin

The primary independent unit is a **root task** sampled from a declared target population. A root can produce several randomized continuations, but those continuations are not independent roots. A user with several related tasks can instead be the independence cluster when cross-task dependence is material.

Let X be the fixed original user request and baseline metadata; let O_0 be the receiver’s initial answer. Set $H_1 = (X, O_0, \text{baseline metadata})$. Turns $t = 1, \dots, T$ are subsequent feedback opportunities. Thus this version estimates feedback effects conditional on the initiating request; it does not estimate the effect of choosing an entirely different original task. To study initial-prompt wording, hold the underlying task and target answer fixed and add an initial randomized wording decision before O_0 .

At turn t :

1. Observe full pre-generation history H_t .
2. Draw an ordered candidate slate $C_t = (a_{t1}, \dots, a_{tK_t})$ from a frozen generator kernel $G_{0t}(dc|H_t)$.
3. Observe all eligible candidates, including an explicit STOP candidate, then draw index J_t from logging selector $b_t(j|H_t, C_t)$.
4. Render and send the exact selected intervention $A_t = a_{tJ_t}$; observe response, costs, and the next history.

The mathematically general action is $U_t = (C_t, J_t)$. Treating this compound action as the intervention avoids hidden assumptions about whether unselected candidates affect cost or future controller behavior. A contrast between j and j' within the same realized slate holds generation fixed and changes the selected prompt.

H_t includes every variable used by the controller or generator, receiver-visible transcript, prior actions, past outputs, available tool results, remaining budget, model versions, and relevant logged prior slate/selection information. It never includes a hidden test label or future evaluator result unavailable at deployment. If previous candidates are neither retained nor used by

any future mechanism, they can be omitted by a declared conditional-independence restriction. A learned embedding is not automatically a sufficient replacement for H_t .

Candidate order, duplicates, truncation, formatting, tool attachments, forbidden-action filters, and rendering are part of treatment definition. The same text at two indices can have different index probabilities. Keep indices and the full slate in the log; do not silently treat index propensity as marginal text propensity.

2.2 Frozen environment and rewards

Freeze receiver model/version, system instruction, decoding parameters, tools, retrieval corpus snapshot where feasible, and evaluator definition. Stochastic receiver responses remain part of the transition distribution. New model versions or externally changing tool state require a new environment stratum or a transport assumption, not a silent pooling operation.

Let R_t be a bounded turn utility, often minus the measured incremental computational cost. Let $Y = y(H_{T+1})$ be terminal quality under a prespecified scoring rule. The total utility is

$$U^{\text{tot}} = \sum_{t=1}^T R_t + Y.$$

One equivalent implementation sets all $R_t = 0$ and includes cumulative costs in terminal Y . Do not count costs twice. Report quality, cost, and stopping time separately even when optimizing $Y - \lambda \text{ cost}$; choose λ before the confirmatory comparison. Resource-constrained optimization and multivariate outcomes are valid alternative targets but are not automatically summarized by a single scalar value.

Generation cost is incurred before selection and is charged even if STOP is then chosen. A cheaper pre-generation stopping gate is a distinct decision stage. A baseline that receives fewer model calls is not cost-matched merely because both systems use the same number of feedback turns.

2.3 STOP, missingness, and failure

STOP accepts the current output. It is an observed treatment that leads to an absorbing state: after STOP, pad the horizon with a deterministic dummy STOP, no further cost, no new receiver call, and an unchanged terminal output. The deterministic padding has likelihood ratio one. Randomized support is required for the actual stop/continue decision, not for artificial padding.

An API failure, missing final score, or lost log is not STOP. A timeout can be a prespecified unfavorable outcome, or it can be censoring requiring a separate observation model. Deleting failed calls estimates a selected-success population. The theorems below assume all defined rewards and terminal outcomes are observed. Missing-at-random and positive observation probabilities can justify added censoring weights; informative missingness is not repaired by treatment weights alone.

3. Policies and estimands

A target policy consists of an externally specified generator G_t^* and selector π_t ; write $d_t(dc, j|h) = G_t^*(dc|h)\pi_t(j|h, c)$. The logging action kernel is $e_t(dc, j|h) = G_{0t}(dc|h)b_t(j|h, c)$. Integrals below include a sum over the finite slate indices.

Main first-stage deployment/evaluation freezes $G_t^* = G_{0t}$ and learns only the selector. Its **marginal text policy** is

$$\bar{\pi}_t(a | h) = \int G_{0t}(dc | h) \sum_j \pi_t(j | h, c) 1\{c_j = a\}.$$

The logged probability $b_t(J_t|H_t, C_t)$ is a conditional selection propensity, not $P(A_t = a|H_t)$ and not a candidate generator token probability.

Define remaining-utility value and action value by a policy-specific Bellman recursion:

$$V_{T+1}^d(h) = y(h), \quad Q_t^d(h, c, j) = E\{R_t + V_{t+1}^d(H_{t+1}) | H_t = h, \text{do}(C_t = c, J_t = j)\},$$

$$W_t^d(h, c) = \sum_j \pi_t(j | h, c) Q_t^d(h, c, j), \quad V_t^d(h) = \int G_t^*(dc | h) W_t^d(h, c), \quad \mathcal{V}(d) = E\{V_1^d(H_1)\}.$$

Every Q depends on the **future** policy. A score trained for generic retry continuation cannot be relabeled as the value under a newly optimized autonomous continuation without updating or reevaluating that continuation.

The within-slate history-specific contrast is

$$\tau_t^d(h, c; j, j') = Q_t^d(h, c, j) - Q_t^d(h, c, j').$$

This answers: among roots at this history and realized slate, what expected downstream utility changes when selecting this candidate instead of the comparator and then using d ? STOP, a specific fixed prompt, or a prespecified stochastic baseline can be the comparator. Historical costs before this decision cancel in the contrast.

A more compact $Q_t^d(h, a)$ requires a **slate-exclusion assumption**: conditional on h and exact rendered a , transition utility and every future policy-relevant variable are invariant to unselected candidates and index. If generation cost is already sunk and subtracted from both compared values, exclusion can hold for the remaining receiver outcome even when it fails for total system cost. If future policies retain slate content, or the receiver sees multiple candidates, exclusion can fail. In those cases retain (h, c, j) .

For a newly arriving user, these are population conditional means for histories like the observed history under the stated environment. Exact-history conditional means are defined almost surely; a unique long transcript typically has no replicated observations. Neural generalization and uncertainty at that transcript require regularity and training support beyond identifiability of a population functional.

4. Identification assumptions

A1 (well-defined intervention and consistency). Selected rendered prompt, slate-generation protocol, STOP, response mechanism, and utility are sufficiently specified. Observed outcomes equal the corresponding potential outcomes for the realized actions. There is no interference across independent root clusters; copying a live state and executing a branch must not mutate another branch’s tools or hidden memory.

A2 (sequential exchangeability). Conditional on full H_t , assignment of U_t is independent of future potential rewards and states under supported interventions. For fixed-generator selection comparisons this can be stated in two steps: slate generation uses only H_t and independent randomness; conditional on (H_t, C_t) , randomized J_t is independent of future potential outcomes. Observational human-feedback logs need all common causes of treatment and outcome measured; latent human knowledge or inaccessible task facts can violate this condition.

A3 (sequential support). At every history reachable under the target, d_t is absolutely continuous with respect to e_t . Ratios must be finite for identification; bounded or sufficiently integrable products are additionally needed for precision and asymptotic inference. Under the same generator the requirement reduces to $\pi_t(j|h, c) > 0 \Rightarrow b_t(j|h, c) > 0$, together with support along the preceding target trajectory. Deterministic target selection is permitted when its selected indices have logging support.

A4 (stable transition and measurement law). Conditional transition kernel $P_t(dr, dh'|h, c, j)$ and scoring rule apply both in logging and target execution. This is not assured by adjusting for a model-name string if the underlying model, tools, evaluator, or task distribution changed.

A5 (sampling and integrability). Root clusters are independent under the primary design; relevant moments exist. Additional boundedness/rate conditions are stated at the result where needed.

These assumptions do not require a low-dimensional Markov state: full history is always the state in this formulation. They also do not make observational confounding an inherent flaw of ordinary Q regression. Correct sequential regression on full sufficient history identifies the same causal Q under A1–A4. A fair comparison includes this adequately adjusted regression, as well as confounded marginal reward prediction.

Theorem 1: longitudinal identification and change of measure

Under A1–A4, the counterfactual law under d is identified by

$$p^d(dh_1, d\bar{u}, d\bar{r}, d\bar{h}') = p(dh_1) \prod_{t=1}^T d_t(du_t | h_t) P_t(dr_t, dh_{t+1} | h_t, u_t). \quad (1)$$

The preceding Bellman equations identify Q_t^d , V_t^d , and $\mathcal{V}(d)$ on supported histories. With

$$\rho_t = \frac{d_t(U_t | H_t)}{e_t(U_t | H_t)}, \quad \Omega_t = \prod_{s=1}^t \rho_s, \quad \Omega_0 = 1,$$

and integrable quantities, the value also satisfies

$$\mathcal{V}(d) = E_e \left[\sum_{t=1}^T \Omega_t R_t + \Omega_T Y \right]. \quad (2)$$

Proof. Consistency identifies the observed transition conditional on a realized treatment with the corresponding potential transition. Sequential exchangeability removes selection into that treatment conditional on prior full history. Apply these facts at the final decision, then induct backward to replace every counterfactual transition by P_t . Integrating the last transition proves the terminal Bellman step; iterating proves (1) and the complete recursion. Absolute continuity implies that the target law through transition t has Radon–Nikodym derivative Ω_t with respect to the logging law through that transition; all transition factors cancel. Therefore $E_d R_t = E_e(\Omega_t R_t)$ and $E_d Y = E_e(\Omega_T Y)$. Sum these equalities. No cross-world joint distribution of two different action outcomes has been used. $\square$

Same-generator corollary. If $G^* = G_0$, then $\rho_t = \pi_t(J_t | H_t, C_t) / b_t(J_t | H_t, C_t)$. The generator probabilities cancel even if a black-box generator cannot provide tractable sequence likelihoods. This does **not** authorize changing that generator while retaining the ratio.

Generator-change corollary. For a different generator with supported, evaluable density ratio,

$$\rho_t = \frac{dG_t^*(\cdot | H_t)}{dG_{0t}(\cdot | H_t)}(C_t) \frac{\pi_t(J_t | H_t, C_t)}{b_t(J_t | H_t, C_t)}. \quad (3)$$

This is the density of the complete ordered slate after its actual sampling, filtering, and deduplication procedure. Multiplying raw token probabilities is correct only if it exactly reproduces that procedure. Without support or a valid generator ratio, collect new randomized trajectories under the revised frozen generator. Updating an actor is an engineering operation; it is not automatically an identifiable offline causal evaluation.

Proposition 2: limits of individual and unsupported-prompt claims

- (a) Randomized data do not generally identify the distribution of literal individual effects $Y_i(a) - Y_i(a')$.
- (b) Without support or structural extrapolation assumptions, the value of an arbitrary unseen prompt is not nonparametrically identified.

Proof. For (a), let the two potential outcomes each be Bernoulli with probability one half. In model I they are identical; in model II they are complements. Independently randomized treatment yields the same observed distribution in both models. Every individual effect is zero in I, whereas effects are -1 or $+1$ in II. Thus the observed law identifies both means and their difference, but not the individual-effect distribution. For (b), choose an action with zero logging probability at a target-relevant history. Two transition kernels can agree on every observed action and assign terminal means zero and one, respectively, to this action. They induce the same logging law and different target values. $\square$

Formal positive token probability for every string is weaker than useful support: probabilities can be so small that finite-sample estimation is effectively unconstrained. Embedding proximity and a language-model prior are extrapolation assumptions, not replacements for this argument.

5. Representations, text treatments, and what a neural critic must learn

Let $S_t = f(H_t)$ and $Z_t = z(A_t)$ be representations. Three distinct claims should not be conflated:

- **Prediction approximation:** a sufficiently flexible $q(S, Z)$ predicts observed outcomes well.
- **Decision sufficiency for a fixed continuation:** $Q_t^d(h, c, j) = q_t^d(f(h), z(c_j))$ for every supported action under the declared exclusion restrictions.
- **Controlled Markov sufficiency:** conditional reward and the next represented state depend on full history and intervention only through the represented state/action; terminal quality is a function of the terminal representation; target/logging policies and generator laws are representable on the same state when a reduced-state likelihood is used.

The second permits estimating the specified Q; the third supports recursive planning and reusable Bellman equations for policies measurable in the representation. Neither follows from a high-dimensional encoder architecture or from predicting average reward accurately.

Proposition 3: sufficient representations and a counterexample

Suppose for all admissible actions the conditional law of (R_t, S_{t+1}) given full history and action depends only on (S_t, Z_t) , terminal Y is measurable in S_{T+1} , slate exclusion holds, and the target generator/selector uses only the represented variables. Then backward recursion gives $Q_t^d(h, a) = q_t^d(S_t, z(a))$.

Proof. At the terminal stage the value is measurable in S_{T+1} . If V_{t+1}^d is a function of S_{t+1} , conditional expectation of $R_t + V_{t+1}^d$ is a function of (S_t, Z_t) by the controlled transition restriction. Integrating over a represented target policy makes V_t^d a function of S_t . Induct backward. $\square$

Without these restrictions, let H be a fair binary variable, A be independently randomized binary treatment, and $Y = 1(A = H)$. An encoder $S = 0$ loses no marginal mean prediction beyond the constant one half, but both encoded action means equal one half; the full-history policy $A = H$ achieves one and an encoded policy achieves at most one half. Thus randomization survives in this example while action-relevant heterogeneity is destroyed. In observational data, collapsing a true confounder can additionally destroy exchangeability.

A prompt embedding $z(a)$ can collapse wording variants with different effects. A well-defined intervention $do(Z = z)$ then requires either treatment-version irrelevance or a specified distribution over exact strings mapping to z . The latter is a stochastic textual intervention, with value depending on that distribution. Decoding an optimized embedding into a new string does not prove either property.

6. Longitudinal doubly robust evaluation

6.1 Target fixed outside the evaluation sample

First freeze d independently of the evaluation sample. This can be a prespecified policy or a policy learned on separate training roots. All nuisance functions below are also fixed when taking conditional expectations; cross-fitting implements this condition fold by fold.

Let $\widehat{Q}_t(h, u)$ be any bounded candidate Q functions and let

$$\widehat{V}_t(h) = \int d_t(du | h) \widehat{Q}_t(h, u), \quad \widehat{V}_{T+1} = Y.$$

Let $\widehat{e}_t$ be positive on target support and define

$$\widehat{\rho}_t = d_t / \widehat{e}_t, \quad \widehat{\Omega}_t = \prod_{s=1}^t \widehat{\rho}_s.$$

The score is

$$D_d(O; \hat{\eta}) = \widehat{V}_1(H_1) + \sum_{t=1}^T \widehat{\Omega}_t \{R_t + \widehat{V}_{t+1}(H_{t+1}) - \widehat{Q}_t(H_t, U_t)\}. \quad (4)$$

A terminal-utility implementation has $R_t = 0$ and $\widehat{V}_{T+1} = Y$ including its accumulated costs. The cross-fitted value estimate averages (4), with every root and all of its branches confined to one fold.

Theorem 4: exact remainder and stagewise robustness

Let $\delta_t = \widehat{Q}_t - Q_t^d$, and assume integrability. Then

$$E_e D_d(O; \hat{\eta}) - \mathcal{V}(d) = \sum_{t=1}^T E_e \left[\widehat{\Omega}_{t-1} \int d_t(du \mid H_t) \left\{ 1 - \frac{e_t(u \mid H_t)}{\widehat{e}_t(u \mid H_t)} \right\} \delta_t(H_t, u) \right]. \quad (5)$$

Thus the score is unbiased when all assignment mechanisms are correct or all target-continuation Q functions are correct. More generally, each stage's summand is zero if **at that stage** either $\widehat{e}_t = e_t$ or $\widehat{Q}_t = Q_t^d$. Arbitrary stagewise choices of these conditions suffice. This statement concerns actual target Q functions, not regression fits that may inherit misspecification from earlier pseudo-outcomes.

Proof. Write $\widehat{V}_t = V_t^d + \delta V_t$ where $\delta V_t(h) = \int d_t \delta_t$, and $\delta V_{T+1} = 0$. For every t , the true Bellman residual $R_t + V_{t+1}^d - Q_t^d$ has conditional mean zero given (H_t, U_t) , hence its product with $\widehat{\Omega}_t$ also has mean zero. Subtract $\mathcal{V}(d)$ from the mean of (4). The remaining expression is

$$E \delta V_1 + \sum_{t=1}^T E \{ \widehat{\Omega}_t \delta V_{t+1} - \widehat{\Omega}_t \delta_t(H_t, U_t) \}.$$

Reindex the first term of the sum, using $\widehat{\Omega}_0 = 1$ and $\delta V_{T+1} = 0$, to obtain

$$\sum_{t=1}^T E \{ \widehat{\Omega}_{t-1} \delta V_t - \widehat{\Omega}_t \delta_t(H_t, U_t) \}.$$

Conditional on H_t , integrate the second term under the true action law e_t : it equals $\widehat{\Omega}_{t-1} \int e_t(d_t/\widehat{e}_t) \delta_t$. Subtraction gives (5). All robustness statements follow by making either factor in each integrand zero. $\square$

For a fixed same generator, use $\widehat{e} = G_0 \widehat{b}$ and $d = G_0 \pi$. Equation (5) reduces to

$$\sum_t E \left[\widehat{\Omega}_{t-1} \int G_{0t}(dc \mid H_t) \sum_j \pi_t(j \mid H_t, c) (1 - b_t/\widehat{b}_t) \delta_t(H_t, c, j) \right]. \quad (6)$$

Known randomization probabilities should ordinarily be used as logged. Estimated propensities cannot fix missing intervention support. Weight clipping changes the exact identity and trades bias for variance; report it as a sensitivity estimator alongside unclipped identification targets.

6.2 Orthogonality and rates

Equation (5) is a product remainder. At truth, a small perturbation in either nuisance while the other stays true has zero first derivative in the score mean. This is the claimed Neyman orthogonality; it is not a guarantee about a neural optimizer finding the correct functions.

For a conditional-on-training statement, define the finite weighted stage measure

$$\mu_t(dh, du) = E_e\{\widehat{\Omega}_{t-1}1(H_t \in dh)\} d_t(du | h).$$

Cauchy–Schwarz applied directly to (5) gives the precise bound

$$|ED_d - \mathcal{V}(d)| \leq \sum_{t=1}^T \left\| 1 - \frac{e_t}{\widehat{e}_t} \right\|_{L_2(\mu_t)} \|\widehat{Q}_t - Q_t^d\|_{L_2(\mu_t)}. \quad (7)$$

The assignment error is **relative** to the fitted assignment density. Bounded likelihood ratios alone do not convert it into an ordinary absolute density-error norm when densities can approach zero. In the fixed-generator finite-selection design, the explicit lower bound $\widehat{b}_t(j|h, c) \geq \varepsilon > 0$ gives $|1 - b_t/\widehat{b}_t| \leq |b_t - \widehat{b}_t|/\varepsilon$; dominated weighted stage measures then give the familiar absolute-propensity-error product bound. Analogous absolute-density claims for unrestricted language require an appropriate lower-bound/domination condition and should not be assumed.

For fixed T , products in (7) that sum to $o_p(n^{-1/2})$, score convergence in L_2 , and independent-root cross-fitting yield asymptotic linearity. Both the relative assignment error and the Q error converging at $o_p(n^{-1/4})$ in these weighted norms is one sufficient regime, not a necessary universal requirement. The mass and domination constants of these measures can grow exponentially with horizon; the bound does not remove the curse of overlap. With known randomized e , the bias remainder is exactly zero even for misspecified Q, although efficiency then depends on Q quality.

Asymptotic expansion. With a fixed finite number of folds, condition on a fold’s training sample. Decompose its score average into the empirical mean of the true score, the centered empirical mean of $D(\widehat{\eta}) - D(\eta_0)$, and its conditional mean. Conditional independence and L_2 score convergence make the centered difference $o_p(n^{-1/2})$ by its conditional variance; (7) controls the mean. Sum the finite folds. The ordinary independent-root central limit theorem then applies to the true influence scores. This proof fails if branches from one root leak across training and evaluation folds.

Theorem 5: efficient influence function in the stated model

Consider a nonparametric finite-horizon full-history model whose unknown factors are the initial-history law, logging action laws, and transition/reward laws. The target action law d is fixed externally and does not vary with this observed-data model. Assume positivity, differentiability along regular submodels, and square-integrability of the expression. Then

$$\phi_d(O) = V_1^d(H_1) - \mathcal{V}(d) + \sum_{t=1}^T \Omega_t \{R_t + V_{t+1}^d(H_{t+1}) - Q_t^d(H_t, U_t)\} \quad (8)$$

is the canonical gradient, hence efficient influence function. It remains canonical if known logging randomization mechanisms are removed from this model.

Proof. A regular submodel score decomposes into $S_0(H_1) + \sum_t S_{e,t}(U_t|H_t) + \sum_t S_{P,t}(R_t, H_{t+1}|H_t, U_t)$, each component having conditional mean zero at its factor. Differentiating the g-formula, the initial-law derivative is $E[(V_1^d - \mathcal{V})S_0]$. The target law does not contain e , so every logging-law derivative is zero. For a transition perturbation at stage t , its derivative is the target expectation of remaining return times $S_{P,t}$. Conditional expectation of the later return is $R_t + V_{t+1}^d$; earlier rewards are measurable before this transition and contribute zero. Change of measure and subtraction of the conditional mean yield $E[\Omega_t(R_t + V_{t+1}^d - Q_t^d)S_{P,t}]$.

Now pair (8) with the whole score. Conditional mean-zero and iterated expectation eliminate all cross-factor terms; only the displayed derivative for the matching factor remains. Also $E\phi_d = 0$. Each summand is a permissible initial or transition tangent component multiplied by a function measurable before its transition. Thus (8) belongs to the tangent-space closure and represents every pathwise derivative, which makes it the canonical gradient. It is orthogonal to all logging-action tangent components; removing known-action score components leaves the same canonical gradient. $\square$

Model boundary. Full history contains previous outcomes, so this result does not exploit time-homogeneous or low-dimensional Markov restrictions. In smaller models the efficient gradient can differ, and marginalized occupancy ratios may be more efficient. Claiming universal efficiency for (8) is incorrect.

6.3 Random slates: fixed design versus natural-generator target

When the generator is an externally fixed sampling protocol G_0 , (4) integrates over this known protocol in each $\widehat{V}_t$; (8) has no generator-score term. Exact integration is a mathematical assumption of these formulas. Finite Monte Carlo slate integration creates additional numerical error; use sufficiently accurate independent draws or carry that error into inference. One should not silently substitute a single logged slate into every integrated value and retain the same efficiency claim.

If instead the estimand keeps the **unknown natural** generator G_P from the observed population, then the target $d(P) = G_P\pi$ itself changes when P changes. In the full-history model the canonical gradient additionally contains

$$\sum_{t=1}^T \Omega_{t-1} \{W_t^{\pi, G_P}(H_t, C_t) - V_t^{\pi, G_P}(H_t)\}. \quad (9)$$

Derivation. A generator perturbation at t contributes its score $S_{G,t}(C_t|H_t)$ to the target g-formula. The remaining conditional target value after seeing C_t is $W_t(H_t, C_t)$. Change the measure only through the previous selection, since the generator distribution is unchanged between logging and target at this turn; this yields $E[\Omega_{t-1}(W_t - V_t)S_{G,t}]$. The conditional mean subtraction makes it a generator tangent component. Adding it to (8) gives the derivative for every model factor. $\square$

Equivalently, treat (H_t, C_t) as the decision state; the usual nonparametric longitudinal score then includes candidate variability. That score can also be used with a known generator as an unbiased but generally less efficient estimator. These are different observation models and efficiency statements, not conflicting algorithms. A fixed finite slate in the simulator makes the extra term identically zero.

7. History-specific causal Q and blip learning

The scalar value score does not by itself give a root-specific label for every prompt. For same-generator randomized slates, define the downstream corrected remaining-return variable

$$Z_{t+1} = \widehat{V}_{t+1}(H_{t+1}) + \sum_{s=t+1}^T \left(\prod_{k=t+1}^s \frac{\pi_k(J_k | H_k, C_k)}{\widehat{b}_k(J_k | H_k, C_k)} \right) \{R_s + \widehat{V}_{s+1}(H_{s+1}) - \widehat{Q}_s(H_s, C_s, J_s)\}, \quad (10)$$

with $Z_{T+1} = Y$. Integration over the fixed generator is understood in $\widehat{V}$. For each candidate index j in the realized slate **whose conditional action value is to be learned**, require both $b_t(j|h, c) > 0$ and $\widehat{b}_t(j|h, c) > 0$ at the relevant histories. This requirement also applies when $\pi_t(j|h, c) = 0$: target-policy support alone does not identify every candidate-specific Q. Excluded or zero-logging-probability candidates receive no such label. For each index meeting these conditions, define

$$\Gamma_{t,j} = \widehat{Q}_t(H_t, C_t, j) + \frac{1\{J_t = j\}}{\widehat{b}_t(j | H_t, C_t)} \{R_t + Z_{t+1} - \widehat{Q}_t(H_t, C_t, j)\}. \quad (11)$$

This is a vector of potentially noisy corrected labels over the supported indices of the finite realized slate. Only one residual is observed, so it is not a full-information counterfactual observation. Fit a critic by honest regression of these labels on (H_t, C_t, j) or on exact prompt text under justified exclusion; use cross-fitting at root level.

Theorem 6: exact conditional bias of the pseudo-outcome

For a queried index satisfying $b_j > 0$ and $\widehat{b}_j > 0$, put $z = (h, c)$, $b_j = b_t(j|z)$, $\widehat{b}_j = \widehat{b}_t(j|z)$, and

$$B_{t+1}(z, j) = E\{Z_{t+1} - V_{t+1}^d(H_{t+1}) \mid H_t = h, C_t = c, J_t = j\}.$$

Then

$$E(\Gamma_{t,j} \mid H_t = h, C_t = c) - Q_t^d(h, c, j) = \left(1 - \frac{b_j}{\widehat{b}_j}\right) \{\widehat{Q}_t(h, c, j) - Q_t^d(h, c, j)\} + \frac{b_j}{\widehat{b}_j} B_{t+1}(z, j). \quad (12)$$

Proof. Conditional on the slate and history, only $J_t = j$ contributes to the residual. Its expected multiplier is $b_j/\hat{b}_j$. By the Bellman equation, $E[R_t + Z_{t+1}|z, J = j] = Q_t^d(z, j) + B_{t+1}(z, j)$. Substitute in (11) and collect terms. $\square$

Consequently, local propensity correctness alone does not remove biased downstream labels. Conditional unbiasedness holds when the future recursion is conditionally unbiased at reachable next histories and the current propensity or current Q is correct. Theorem 4 applied starting from any such history supplies sufficient downstream conditions. Contrasts $\Gamma_{t,j} - \Gamma_{t,j'}$ estimate the conditional blip after regression under the same conditions.

Weighted squared loss, contrast losses, or ranking losses can train a neural critic, but they solve different statistical problems. Squared regression targets Q calibration; direct contrast loss can target heterogeneity; ranking loss targets decisions and need not calibrate Q. Bootstrap ensemble spread alone is not a proved pointwise confidence interval for the conditional causal effect. Smoothness, complexity, overlap, and distributional calibration assumptions are needed for honest conditional inference. With continuously varying histories or text, a point-evaluation Q is generally not a regular root- n functional in the unrestricted model; the root- n result for marginal policy value does not transfer to every neural prediction.

8. Optimality, policy improvement, and conservative selection

8.1 Supported optimal policy

For a fixed generator G , let $V_t^{*,G}$ solve

$$Q_t^{*,G}(h, c, j) = E\{R_t + V_{t+1}^{*,G}(H_{t+1}) \mid h, c, j\}, \quad V_t^{*,G}(h) = E_{C \sim G(\cdot|h)} \max_{j \in \mathcal{J}(h,C)} Q_t^{*,G}(h, C, j). \quad (13)$$

$\mathcal{J}$ contains eligible, supported candidates including STOP. With finite nonempty slates and measurable tie breaking, backward induction gives an optimal deterministic selector within this generator-constrained policy class. The maximization occurs after seeing the slate, so $E \max_j Q$ is appropriate; $\max_j E Q$ is generally not.

Proof. At the last decision, conditional expected utility is maximized by the largest supported Q. Assuming optimal continuation values are available at $t + 1$, the same pointwise maximization at t dominates every other first-stage selection followed by any admissible continuation. Integrating over the generator and inducting backward proves global optimality within the class. $\square$

Theorem 7: performance difference and certified local improvement

Let β and π be selectors sharing the same generator and environment. Then

$$\mathcal{V}(\pi) - \mathcal{V}(\beta) = \sum_{t=1}^T E_{\pi,G} \left[\sum_j \{ \pi_t(j \mid H_t, C_t) - \beta_t(j \mid H_t, C_t) \} Q_t^\beta(H_t, C_t, j) \right]. \quad (14)$$

If a simultaneous event guarantees $|\widehat{Q}_t^\beta(h, c, j) - Q_t^\beta(h, c, j)| \leq u_t(h, c, j)$ on all target-reachable histories/candidates, define

$$L_t(h, c; \pi, \beta) = \sum_j (\pi_j - \beta_j) \widehat{Q}_t^\beta(h, c, j) - \sum_j |\pi_j - \beta_j| u_t(h, c, j). \quad (15)$$

On that event, $\mathcal{V}(\pi) - \mathcal{V}(\beta) \geq \sum_t E_{\pi,G} L_t$. In particular, restricting policy changes to those with $L_t \geq 0$ pointwise guarantees nondecrease on the same event.

Proof. Under target trajectories sum the baseline Bellman residuals $R_t + V_{t+1}^\beta(H_{t+1}) - V_t^\beta(H_t)$. The state values telescope to $U^{tot} - V_1^\beta(H_1)$. Conditional averaging gives $E_G \sum_j \pi_j Q_t^\beta - V_t^\beta = E_G \sum_j (\pi_j - \beta_j) Q_t^\beta$, proving (14). Substituting $Q = \widehat{Q} + (Q - \widehat{Q})$ and bounding each error term below by $-|\pi_j - \beta_j| u_j$ proves (15) and its consequences. $\square$

The required simultaneous event is a substantial statistical condition. Ordinary per-candidate pointwise intervals, post hoc best-of-many selection, or uncalibrated neural uncertainty do not establish it. An average advantage on logging histories alone also

does not prove improvement under a changed trajectory distribution. A safer practical release gate is an independent root-level evaluation of the frozen proposed policy against baseline, with uncertainty for the policy-value difference.

STOP is handled by the same theorem. A deterministic continue candidate is preferred to STOP only if its supported downstream utility exceeds the stopping value after all future costs. The guarantee applies to the measured utility, not to an unmeasured preference or an oracle label leaked into policy input.

8.2 Finite-policy holdout guarantee

Suppose a training sample fixes m proposed policies and their nuisance functions. An independent evaluation set contains n independent roots, and a score for each policy lies in $[a, b]$ almost surely. If assignment probabilities are correct, its score mean is unbiased. Then, with probability at least $1 - \alpha$, simultaneously for every policy,

$$|\widehat{\mathcal{V}}(d) - \mathcal{V}(d)| \leq (b - a) \sqrt{\frac{\log(2m/\alpha)}{2n}}. \quad (16)$$

Proof. Apply Hoeffding’s inequality to each policy’s independent bounded root scores and union bound over m . If only a known bias bound r_d from (5) is available, add it to that policy’s radius. $\square$

A policy can be released when its lower value bound exceeds the baseline upper bound, or more efficiently when a paired score-difference lower bound exceeds zero. This is a finite-class holdout result; repeatedly inspecting the same holdout and generating additional policies violates its fixed-class premise. Fresh holdouts, a predeclared simultaneous class, or sequentially valid inference are needed for ongoing adaptation.

8.3 Candidate coverage versus critic error

To compare with unrestricted prompts, additionally assume slate exclusion, candidate-independent action reward/transition, a common cost convention, and an admissible prompt space $\mathcal{A}(h)$ with a measurable maximum or arbitrarily close maximizers. Let Q_t^*, V_t^* denote unrestricted optimal continuation values. Every slate maximum, greedy selection, and error bound in this subsection is over the **same eligible, positive-logging-support index set** $\mathcal{J}(h, c)$ used in (13), which is nonempty and contains STOP. Every compared selector assigns zero probability outside this set. Merely appearing in the slate does not make an excluded or unsupported candidate selectable. Define the candidate coverage loss

$$c_t(h) = V_t^*(h) - E_{C \sim G} \max_{j \in \mathcal{J}(h, C)} Q_t^*(h, C_j) \geq 0. \quad (17)$$

This is a theoretical quantity; the global optimal prompt over all language is usually unknown. It must not be estimated by labeling the best sampled candidate globally optimal.

Theorem 8: coverage decomposition and finite-horizon bounds

For any candidate selector $\hat{\pi}$ sharing G ,

$$\mathcal{V}^* - \mathcal{V}(\hat{\pi}, G) = \sum_t E_{\hat{\pi}, G} c_t(H_t) + \sum_t E_{\hat{\pi}, G} \left[\max_{j \in \mathcal{J}(H_t, C_t)} Q_t^*(H_t, C_{tj}) - \sum_{j \in \mathcal{J}(H_t, C_t)} \hat{\pi}_t(j | H_t, C_t) Q_t^*(H_t, C_{tj}) \right]. \quad (18)$$

If the critic estimates the **restricted optimal** $Q_t^{*,G}$ with uniform error at most ε_t over every relevant history, slate, and index in $\mathcal{J}(h, c)$, and $\hat{\pi}$ greedily selects its largest prediction **within that same set**, then

$$\mathcal{V}^* - \mathcal{V}(\hat{\pi}, G) \leq \sum_t \sup_h c_t(h) + 2 \sum_t \varepsilon_t. \quad (19)$$

Proof. Telescope unrestricted optimal state values along trajectories of $\hat{\pi}$: $\mathcal{V}^* - \mathcal{V}(\hat{\pi}) = \sum_t E[V_t^*(H_t) - Q_t^*(H_t, A_t)]$. Add and subtract the best eligible supported slate value; conditioning on pre-slate history yields (18). Every maximum is over $\mathcal{J}$, so an unselectable high-value candidate cannot eliminate the coverage loss.

For (19), first compare V_t^* with $V_t^{*,G}$. Their Bellman equations and $\max x - \max y \leq \max(x - y)$ imply

$$\sup_h (V_t^* - V_t^{*,G}) \leq \sup_h c_t(h) + \sup_h (V_{t+1}^* - V_{t+1}^{*,G}).$$

The terminal difference is zero, hence the unrestricted-versus-restricted gap is at most the sum of coverage suprema. Next, for any slate, let j^* maximize the true restricted optimal Q over $\mathcal{J}(h, c)$ and $\hat{j}$ maximize its estimate over that same set. Then

$$Q^{*,G}(j^*) - Q^{*,G}(\hat{j}) \leq |Q^{*,G}(j^*) - \widehat{Q}(j^*)| + |\widehat{Q}(\hat{j}) - Q^{*,G}(\hat{j})| \leq 2\varepsilon_t,$$

because $\widehat{Q}(j^*) \leq \widehat{Q}(\hat{j})$. Telescope $V^{*,G}$ along the greedy estimated policy; each conditional stage gap is at most $2\varepsilon_t$. Add the two bounds. $\square$

Do not apply (19) to a critic of an arbitrary baseline continuation. Such a critic supports Theorem 7’s improvement analysis, not a global near-optimality claim.

Coverage corollary. Suppose a protocol starts with K independent proposals conditional on history. Each proposal has probability at least $p_t(h)$ of simultaneously (i) having $Q_t^*(h, A) \geq V_t^*(h) - \eta_t$ and (ii) meeting a proposal-wise admissibility criterion that guarantees at least one corresponding index remains eligible and selectable with positive logging probability after the complete protocol. Assume $0 \leq \eta_t \leq B_t$, where B_t bounds the Q range. Then

$$c_t(h) \leq \eta_t + B_t \{1 - p_t(h)\}^K. \quad (20)$$

Proof. The qualifying near-optimal-proposal events are independent because they are functions of independent original draws and the fixed history. The probability of none is at most $(1 - p)^K$. If at least one occurs, the preservation condition puts an η_t -optimal index in the final selectable set $\mathcal{J}$, so the gap is at most η_t ; otherwise it is at most B_t . If r is the actual failure probability, the expected gap is at most $\eta_t + (B_t - \eta_t)r$. Since $0 \leq \eta_t \leq B_t$ and $r \leq (1 - p)^K$, this is at most $\eta_t[1 - (1 - p)^K] + B_t(1 - p)^K$, which implies (20). $\square$

Here K counts the **original independent draws**, not a random post-filter count or the final number of unique candidates. Deduplication that preserves one selectable copy of every qualifying proposal preserves the maximum event and does not invalidate the bound. Correlated beams, globally dependent eligibility filtering, or removing qualifying candidates can invalidate the independent-event or preservation premises; derive their coverage separately. An unsupported candidate has no qualifying probability for this corollary regardless of its oracle quality. More proposals can raise generation cost, so (20) for quality alone does not establish improved net utility as K grows.

9. Training a generative policy and a dynamic skill

An implementable cycle is: freeze generator and receiver; generate and randomize supported slates; fit Q/blip models; learn a selector on training roots; evaluate the frozen selector on new roots; only then use high-value evidence to update the generator and repeat collection. The learned controller is a dynamic skill only in the operational sense of an adaptive policy with a declared model/environment scope.

For a differentiable stochastic generator-selector law d_θ , with parameter-independent transition law and dominated differentiation, the policy-gradient identity is

$$\nabla_\theta \mathcal{V}(d_\theta) = E_{d_\theta} \sum_{t=1}^T \nabla_\theta \log d_{\theta t}(U_t | H_t) Q_t^{d_\theta}(H_t, U_t). \quad (21)$$

Derivation. Differentiate the trajectory density product to obtain a sum of action log-density derivatives multiplied by total return. For a stage’s score, rewards accumulated before that action have zero covariance with it because the conditional score

mean is zero. Conditional expectation of subsequent return given history and action is $Q_t^{d_\theta}$. Substitute to get (21). Any history-only baseline can be subtracted by the same zero-mean argument. $\square$

The identity uses on-policy history occupancy and the current policy’s true Q. Maximizing $E_{H \sim \text{old logs}, A \sim G_\theta} \widehat{Q}(H, A)$ is generally a surrogate; it omits changing history occupancy and can exploit critic error. Off-policy gradients require valid full likelihood ratios/occupancy correction and appropriate nuisance control. Distilling selected prompts, reward-weighted fine-tuning, direct preference objectives, and KL regularization are proposed optimization methods, not proved substitutes for (21) or randomized final evaluation.

A learned transition/world model can support planning only under identified transitions and sufficient state representation. Rollouts generated by that model are model predictions, not fresh counterfactual validation data. Longer simulated planning compounds transition error and can reward unrealistic states. Keep synthetic-world evaluation separate from actual frozen-receiver executions.

10. Honest inference, branch experiments, and transport

10.1 Shared prefixes and independent roots

A forked experiment restores the same declared receiver/tool snapshot and executes candidate continuations. Repeated rollouts estimate the conditional mean at that snapshot over receiver and continuation randomness. They do not reveal a unique user’s latent joint counterfactual outcomes. If shared random seeds are used, the coupling can reduce variance but is an experimental design choice; it is not the naturally occurring cross-world coupling.

All continuations sharing root task, common prefix, user, template, or stochastic cached output must stay in the same independence cluster when those shared variables create dependence relevant to the target. Fit nuisance models and select policies on other clusters. Reporting the total number of branches as n overstates independent information.

For m independent roots with a prespecified within-root averaging rule, let S_i be the root’s averaged evaluation score, including any valid branch-selection correction. The root-average estimand is estimated by $\widehat{V} = m^{-1} \sum_i S_i$, with

$$\widehat{\text{Var}}(\widehat{V}) = \frac{1}{m(m-1)} \sum_{i=1}^m (S_i - \bar{S})^2. \quad (22)$$

Justification. Independence applies to root scores, not branch scores. The sample variance of independent root scores is unbiased for their variance when roots are identically sampled; dividing it by m gives (22). A root-level central limit theorem or root bootstrap supports large-sample inference. Unequal/informative root sampling, adaptive branch allocation, or branch pruning needs explicit design weights; a convenience mean across all branches targets a branch-size-weighted distribution.

Branches must preserve the full system state. Restoring only visible text while tool files, caches, hidden conversation memory, or server model versions change violates the branch experiment’s consistency assumption. A behavioral simulator supplies exact counterfactuals only within its own specified simulator law.

10.2 Learned policies and nonregular optima

A conditional interval for a policy frozen after training is different from an interval for the population-optimal value. If different folds train different policies, the straightforward cross-fitted estimator targets an average of fold-trained policy values; it does not automatically evaluate the final all-data-trained policy. Evaluate that final policy on fresh roots. At ties or near ties, the maximum-value functional can be nonregular. Ordinary plug-in Wald intervals for an estimated argmax require extra conditions; the fixed-policy results above do not supply them.

Calibration reports should separately assess policy-value error, supported candidate ranking, blip sign, conditional mean calibration across predeclared history groups, uncertainty coverage, and realized deployment utility. A single global value interval does not establish individualized calibration.

10.3 New users and environment transport

A new user from the same target population is covered only to the extent that the relevant history is supported. For a different baseline-history distribution $q(H_1)$ with unchanged conditional mechanisms, known or estimable baseline density ratio $w = q/p$ can transport the value by weighting the initial distribution and scores. New receiver versions or different latent user preferences change conditional mechanisms and require stronger assumptions or new experiments. Logging version metadata enables auditing; it does not by itself prove invariance.

10.4 Historical logs versus prospective trials

Historical logs are useful for initialization, candidate proposal, and hypothesis development. Their causal use depends on measured confounders, treatment versions, stopping/missingness definitions, and support. A deliberately confounded simulation can demonstrate failure of a marginal reward model, but it should also include correctly adjusted sequential regression as a comparator. Prospective within-slate randomization removes selection confounding at that decision by design, provided the receiver continuation and outcome recording obey the protocol.

11. Research claims that remain conditional or open

1. **New free-form prompt performance:** not identified by the finite-slate randomization alone. Needs covered generation, structural generalization, or fresh trials.
2. **Uniform neural confidence bands:** assumed in Theorem 7, not constructed or validated here. Independent policy-level holdouts are the immediate defensible alternative.
3. **Universal representation sufficiency:** not established. Evaluate history/text compression through controlled ablations and conditional calibration; failure to detect bias does not prove sufficiency.
4. **Generator improvement after distillation:** requires new frozen-generator evaluation or valid full-generator ratios. Winner distillation can amplify noise and erase exploration.
5. **Prompt-level causal attribution to latent semantic factors:** requires explicitly defined semantic interventions and treatment-version assumptions. A latent vector coordinate is not automatically a manipulable treatment.
6. **Unmeasured confounding in human logs:** randomized selection evidence does not retroactively identify unsupported historical interventions. Sensitivity analysis can quantify assumptions but is not a proof they hold.
7. **Human personalization and preferences:** one frozen automatic evaluator estimates that evaluator’s outcome; it does not establish a user’s satisfaction, trust, or welfare.
8. **Joint receiver/generator adaptation:** outside the frozen-environment theorems unless adaptation state and intervention protocol are modeled. Changing both players is a different target.
9. **Multivariate win outcomes:** can be studied with a specified pairwise functional and coupled/independent policy draws, but do not inherit the scalar Bellman recursion without a defined scalar utility or augmented-state target.
10. **Novelty and practical superiority:** no theorem here establishes that the combined method is new or outperforms current prompt optimizers. Those are literature and empirical questions.

12. Source grounding

The proofs in this document are supplied explicitly for this project’s notation and assumptions; the underlying longitudinal DR and efficiency principles are established. Primary sources used to check those boundaries:

- Jiang, N. and Li, L. (2016). [Doubly Robust Off-policy Value Evaluation for Reinforcement Learning](#). Sequential DR evaluation and its policy-evaluation context.
- Bang, H. and Robins, J. M. (2005). [Doubly Robust Estimation in Missing Data and Causal Inference Models](#). Longitudinal and missing-data DR foundations; [author-hosted full text](#).
- Kallus, N. and Uehara, M. (2020). [Double Reinforcement Learning in Markov Decision Processes](#). Distinguishes efficiency models, non-Markov history ratios, and Markov structure. This project does not claim every history-weighted estimator is efficient under Markov restrictions.
- Luedtke, A. R. and van der Laan, M. J. (2016). [Statistical inference for the mean outcome under a possibly non-unique optimal treatment strategy](#). Nonregularity and inference for optimal versus learned policies.

- van der Laan, M. J. and Luedtke, A. R. (2015). [Targeted Learning of the Mean Outcome under an Optimal Dynamic Treatment Rule](#). Dynamic-rule learning and data-adaptive targets.

The attachment's recent paper titles, venue-status claims, and claims that a research area is unoccupied are research leads, not assumptions or validated novelty evidence for these theorems. The separate literature audit controls those statements.

Stopping, prefix replay, and candidate selection: a scoped theory addendum

Date: 2026-09-20. **Status:** proved under the conditions stated below; exact finite-state checks supplied separately. This addendum supplements [the original theory](#) without replacing its general adaptive-prompt estimand. It reconciles the narrower full-rollout stopping proposal in the experiments draft. Classical change-of-measure and optimal-stopping arguments are not claimed as new foundational results.

1. The decision that an unweighted full rollout actually evaluates

Fix a receiver, generator, evaluator, task population, and maximum of D continuation decisions. Index artifacts by $k=0, \dots, D$: artifact zero is the initial receiver answer, and action A_k produces artifact $k+1$. This avoids mixing artifact numbers with feedback decisions.

A full rollout always executes the declared continuation kernel $e_k(da|H_k)$ at every decision, including histories at which a candidate stopping policy would have stopped. Let H_k contain the full recorded prefix, Y_k the analyst’s score of the current artifact, and C_k the cost that accepting that prefix would have incurred. Define the prefix utility

$$U_k = Y_k - \lambda C_k, \quad 0 \leq k \leq D.$$

C_k includes the initial call and all calls, checks, and generation already performed before that stop decision, using one declared convention. It excludes calls made later solely for full-rollout data collection. Those later calls are real research expenditure and must still be recorded separately. Prefix replay cannot make an already incurred experimental bill disappear.

Let $\mathcal{F}_k^O$ be the sigma-field generated by **all public observations through decision k** , including the information actually available when deciding to stop. Let $\mathcal{F}_k^A$ additionally contain analyst-only labels recorded by that time. The public fields exclude hidden tests and their derivatives. $\mathcal{F}_k^O \subseteq \mathcal{F}_k^A$, and both are increasing filtrations. Writing $\sigma(\varphi_k(H_k))$ alone does not ensure a filtration when the feature map forgets previous information; use $\sigma(\varphi_0(H_0), \dots, \varphi_k(H_k))$, or explicitly restrict a policy to a memoryless feature class.

A frozen stopping rule $\tau \in \{0, \dots, D\}$ is a stopping time for the public filtration: $\{\tau = k\} \in \mathcal{F}_k^O$. Independent policy randomness can be sampled before the rollout and added to the public filtration. “Frozen” means the rule, feature map, thresholds, and hyperparameters are fixed independently of the evaluation roots. Training on other roots is allowed; conditioning on that training data treats the rule as fixed.

Assumptions for prefix replay

S1 — Matching continuation law. Before stopping, the target policy uses exactly the full-rollout continuation kernel e , including generator/class probabilities, within-class text generation, filters, rendering, receiver and tool kernels. Matching a class name while changing the prompt generator is not matching the law. The target continuation itself must be implementable from its allowed information.

S2 — Prefix consistency and no anticipation. Executing additional calls after a prefix does not change the artifact, measured outcome, or charged prefix cost already associated with that prefix. Before stopping, the target and collection protocol induce the same conditional distribution of prefixes. A disclosed future horizon, shared mutable files, different truncation, batch contention affecting the target metric, or a receiver instructed that it must complete all rounds can invalidate this condition.

S3 — Stop outcome is the recorded prefix functional. Stopping means accepting the current artifact, with utility exactly U_k when stopping at k . It does not trigger a final rewrite, a cleanup tool, a new judge-dependent selection, deterioration before a fixed later endpoint, or another unrecorded event. Any such effect must be added to the target and measured. A repeatable scorer alone does not establish S2 or S3.

S4 — Complete observation for the target horizon. Every needed prefix’s public state, grade, and cost is recorded, and all failures have a prespecified measured meaning. The administrative cap is shared by target and collection. Missing continuations or unavailable scores are handled under Section 4 rather than discarded.

S5 — Honest rule and root sampling. Rules use only decision-time public information and are frozen outside the evaluation sample. Independent roots, or a declared independent clustering unit, are sampled from the target distribution. A task-weighted target and a family-weighted target are different if family sizes differ.

These are assumptions about a particular data design. They are not consequences of calling STOP an action or of a deterministic verifier.

Theorem S1: unweighted full-rollout evaluation

Under S1–S4, for any frozen public stopping time τ , the distribution of utility from actually running e until τ and accepting that artifact equals the distribution of U_τ read from a full rollout under e . Therefore

$$V(e, \tau) = E_e[U_\tau] = E_e \left[\sum_{k=0}^D 1\{\tau = k\} U_k \right]. \quad (S1)$$

Under S5, the root-average estimator is unbiased conditional on training and requires no treatment or stopping likelihood ratios: $W=1$.

Proof. Couple target and full-rollout executions using the same initial state and the same exogenous random draws for each continuation, receiver, and tool call. S1 and S2 imply by induction that the two histories are identical through any time before the target stops. Since τ depends only on the public prefix and its presampled randomization, both executions first satisfy the stopping condition at the same prefix. S3 makes the target utility equal to the full rollout’s recorded U_τ pathwise. S4 makes that quantity observed. Integrating proves the equality in distribution and (S1). Independent root sampling and a rule fixed outside evaluation yield unbiased averaging. The additional trajectory suffix is unused. $\square$

What “STOP needs no randomization” means. Under S3, its outcome is an observed prefix functional. No action-assignment probability for STOP is needed to identify that functional. If the logger never selects STOP, its STOP probability is zero: ordinary treatment positivity for STOP does **not** “hold by degeneracy.” Instead a structural observation assumption bypasses the missing STOP action outcome. This is compatible with the original theory’s action-based positivity theorem: that theorem did not assume every stop outcome was independently observed along a full continuation.

What is not necessary. No independence between prefix grades is required; they may be nonstationary and strongly dependent. Optional stopping does not bias an ordinary sample mean of a bounded, fixed, fully observed path functional. No particular critic architecture is needed. A fixed deterministic grader is convenient but not logically necessary if the target is the same recorded stochastic measurement and measurement randomness is included in the declared data law. It is insufficient to infer a different latent quality or a fresh post-stop noisy measurement without an additional calibration/measurement assumption.

What is not implied. The empirical best rule chosen on these same roots has optimistic in-sample value. Complete observation eliminates missing action outcomes for this restricted stopping problem; it does not eliminate overfitting, make hidden scores deployable, establish precision, or prove that a learned policy beats a strong comparator.

2. When weights return: a changed continuation policy

Let $\pi_k(da|H_k)$ be a different, supported continuation law, and let τ remain a public stopping time. Keep S2–S4 and a common baseline distribution. Define

$$W_0 = 1, \quad W_k = \prod_{s=0}^{k-1} \frac{\pi_s(A_s | H_s)}{e_s(A_s | H_s)}.$$

Only actions executed **before** accepting artifact k enter W_k ; there is no STOP factor.

Theorem S2: stopped-prefix change of measure

Under sequential support and invariant transition kernels,

$$V(\pi, \tau) = \sum_{k=0}^D E_e[1\{\tau = k\} W_k U_k] = E_e[W_\tau U_\tau]. \quad (S2)$$

Proof. The event $\{\tau = k\}$ and U_k are measurable in the prefix through k (including measurement variables). The likelihood ratio for that prefix under π versus e is W_k , since receiver and measurement kernels cancel. Change of measure separately for each event yields $E_\pi[1\{\tau = k\}U_k] = E_e[1\{\tau = k\}W_k U_k]$. Sum the finitely many events. This also proves $E_e W_\tau = 1$ by setting every $U_k=1$. $\square$

If $\pi = e$ on all reachable pre-stop prefixes, $W_\tau = 1$. Conversely a generic claim of unweighted validity for every bounded prefix outcome requires equality of the corresponding stopped-prefix laws: otherwise the indicator of a measurable set where the laws differ provides a counterexample. For one specific utility, different laws can accidentally have the same mean; thus “only matching continuation” is a sufficient design guarantee and, up to distributional equivalence, necessary for universal replay validity, not an assertion that coincident means are impossible.

Elementary counterexample. One continuation draws $A \sim \text{Bernoulli}(1/2)$, with final grade $Y_1=A$, and $\tau = 1$. Replay gives one half. A target that always chooses $A=1$ has value one. Its correct ratio is $2 \setminus, 1 \setminus \{A=1\}$. Seeing all prefix grades does not reveal the receiver response to the untaken continuation action.

Diagnostic-arm conflict. If full rollout chooses uniformly among eight classes including three intentionally harmful/privileged diagnostics, a matching stopping policy inherits those diagnostics whenever it continues. It is not in a policy class that forbids those classes. To evaluate a fair stopping-only policy without weights, collect full rollouts under a separately frozen admissible continuation law (for example the declared baseline’s allowed mixture), or use genuinely matching existing rollouts. Dropping diagnostic branches does not automatically produce an unweighted sample of full trajectories from that admissible law. It changes selection probabilities and can change history distribution. Stopping immediately is a trivial exception; no continuation is then used.

3. What the policy can know: stopping values and the oracle gap

The analyst knows U_k for a realized prefix. The deployable rule usually does not. Define

$$s_k = E[U_k \mid \mathcal{F}_k^O].$$

This is the conditional utility of accepting now. Predicting it from public information is a supervised conditional-mean problem. The continuation alternative remains a sequential problem: its value depends on future actions, public information, receiver transitions, and costs.

Theorem S3: public optimal stopping recursion

Under a fixed continuation law and bounded integrable prefix utilities, define the public Snell recursion

$$S_D = s_D, \quad c_k = E[S_{k+1} \mid \mathcal{F}_k^O], \quad S_k = \max(s_k, c_k), \quad k < D. \quad (\text{S3})$$

Then $E S_0$ is the largest achievable expected utility over public stopping times. The first time $s_k \geq c_k$ (with mandatory stop at D) is optimal.

Proof. For any public stopping time, $EU_\tau = \sum_k E[1\{\tau = k\}E(U_k \mid \mathcal{F}_k^O)] = E s_\tau$. At D , accepting is mandatory. At an earlier prefix, stopping gives s_k ; continuing and then behaving optimally gives $E[S_{k+1} \mid \mathcal{F}_k^O]$ by induction. The greater is attainable by an $\mathcal{F}_k^O$ -measurable choice. Backward induction gives both the upper bound and an attaining stopping rule. $\square$

Measurability versus computational access. If a benchmark and its hidden tests are fixed deterministic functions of the task and artifact, the grade may mathematically be measurable in the sigma-field generated by the full task/artifact even though the deployed program is forbidden to run those tests or cannot compute that function. Removing the grade column alone does not prove a strict sigma-field separation. A formal information model can treat the withheld evaluation assets as latent random baseline variables; an operational implementation can instead restrict the allowed feature computations and policy class explicitly. The Snell optimum assumes knowledge of the conditional law and is a theoretical class optimum, not a claim that a fitted model can compute it. The public/analyst distinction must use the declared interpretation consistently.

This recursion is on the **entire public filtration**, not automatically on an 87-coordinate current feature vector. A feature-state implementation needs conditional value/transition sufficiency, or must be presented as a restricted approximation. Conversely, $\mathcal{F}_k^O$ being strictly smaller than the analyst’s information does **not** itself prove insufficiency: excluded hidden variables may

be irrelevant to the conditional law of all future public outcomes. The original theory's representation assumptions remain the correct criterion.

Theorem S4: regret from stopping-advantage error

Let $\Delta_k = c_k - s_k$ be the optimal public continuation advantage. A fitted rule $\hat{\tau}$ continues when $\widehat{\Delta}_k > 0$ and stops otherwise. For the same fixed continuation law, define $r_k = |\Delta_k|$ if its current stop/continue choice disagrees with the optimal sign, and zero otherwise. Then

$$V^* - V(\hat{\tau}) = \sum_{k=0}^{D-1} E[1\{\hat{\tau} \geq k\} r_k] \leq \sum_{k=0}^{D-1} E[1\{\hat{\tau} \geq k\} |\widehat{\Delta}_k - \Delta_k|]. \quad (\text{S4})$$

If $|\widehat{\Delta}_k - \Delta_k| \leq \varepsilon_k$ almost surely and $P(|\Delta_k| \leq x) \leq C_k x^{\alpha_k}$ for $x > 0$, the regret is at most $\sum_k C_k \varepsilon_k^{1+\alpha_k}$.

Proof. At a reached decision, the difference between the optimal public value S_k and the fitted action evaluated with optimal continuation is zero for an optimal choice and $|\Delta_k|$ for the wrong choice. Telescope these one-step Bellman differences along the fitted stopped path; the terminal comparison vanishes, giving the equality. A sign disagreement implies $|\Delta_k| \leq |\widehat{\Delta}_k - \Delta_k|$, proving the first bound. Uniform error confines every mistaken decision to $|\Delta_k| \leq \varepsilon_k$; bound its regret by ε_k and apply the margin probability bound. Dropping the reached-event indicator only enlarges the bound. $\square$

The theorem estimates the **combined** continuation-minus-stop advantage. Estimating only $P(\text{current artifact correct} | \text{public features})$ does not identify continuation gains or the best next feedback. The bounds are conditional on error/margin control; they do not assert those conditions hold for the current fitted critic.

Proposition S5: observable oracle ceilings and unavailable information

For any public stopping rule satisfying S1–S4,

$$E[U_\tau] \leq E\left[\max_{0 \leq k \leq D} U_k\right]. \quad (\text{S5})$$

Proof. $U_\tau \leq \max_k U_k$ for each complete rollout. Integrate. $\square$

The right side is a retrospective full-path upper bound, not a deployable rule. Even revealing the **current** hidden grade is extra information; revealing all future grades is additionally anticipatory. A pathwise maximum of net utility is at least as large as every causal stopping value but need not be attainable even by an analyst who sees only current hidden grades. For independent fair binary grades with no informative public signal and zero cost, any allowed rule has mean one half, while the best of two hidden grades has mean three quarters. Thus an oracle gap need not be capturable by any policy in the allowed information set.

The inequality covers only the same complete continuation law and recorded horizon. It is not an upper bound on another prompt policy or on continuing past an observed natural dropout. If a log ended early and the proposed “oracle” imputes a repair rate for missing suffixes, that number is a **model-based extrapolation**; it is not the observed right side of (S5). The reported +0.046 from the sibling audit explicitly makes such an imputation and must retain that qualifier.

4. Failure, time caps, and incomplete observation

If a service error occurs during continuation, define before collection whether it is a transition to an absorbing failure artifact with a specified utility or a missing observation. Under the first choice it is an observed outcome; full-rollout replay stays valid only if the target incurs the same error law and all prefix utilities remain defined. A collected timeout cannot be silently removed because it weakens the policy.

For the second choice, let R_k indicate that the complete information needed to evaluate prefix k is observed, with monotone missingness. Let g_s be the conditional probability of observing the next prefix, given recorded past information and $R_s=1$. Assume sequential coarsening at random: conditional on that past, survival of observation is independent of future complete prefix variables. Assume $g_s > 0$ wherever the target can continue, and put $G_k = \prod_{s < k} g_s$, $G_0=1$. Scores missing at the initial prefix require their own observation factor.

Theorem S6: censoring-weighted prefix evaluation

Under the preceding observation assumptions and S1–S3,

$$V(e, \tau) = \sum_{k=0}^D E \left[\frac{R_k}{G_k} 1_{\{\tau = k\}} U_k \right]. \quad (\text{S6})$$

The summand is evaluated only when $R_k=1$, in which case its whole required prefix is available.

Proof. For any integrable complete-prefix variable F_k , iterate conditional expectation over the observation indicators. At each observation decision, the conditional mean of its factor $1_{\{R_{s+1}=1\}}/g_s$, given the complete history relevant to F_k and prior observed survival, is one by coarsening at random. Thus $E[R_k F_k/G_k] = E F_k$. Apply this with $F_k = 1_{\{\tau = k\}} U_k$ and sum. $\square$

A different target continuation adds its action ratios from (S2). If a target would continue beyond a deterministic wall-time cap where $g_s=0$, the future value is not identified without further modeling. A full nominal turn budget does not imply complete observation when slow or failing trajectories are truncated. Conditioning on successful completion generally changes the estimand. If a common wall-time deadline itself is part of the policy, define it as an observed intervention/outcome rule with complete utility, and ensure prefix costs/latencies remain transferable under S2.

5. Independent resampling and candidate selection are different interventions

Suppose a bank contains candidate outputs $B_1, \dots, B_K$ generated independently conditional on the task by a fixed initial-prompt receiver law. A selector sees allowed candidate information $\mathcal{G}$, then chooses index J ; hidden candidate scores $Y_1, \dots, Y_K$ are available to the analyst. When all candidates and grades are observed, a frozen selector’s value is $E Y_J$ without action weights. This is **full-information selection from a fixed candidate bank**, not identification of the effect of a new feedback message at a changing conversation history.

Independence is useful for modeling resampling but not needed for the unweighted selection identity. What is needed is that deployment produces the same joint bank distribution, allowed signals, and costs. A bank made from different model versions, adaptive routing, or multi-round continuations defines a different selection target. Exchangeability among repeated samples does not turn them into independent task clusters.

Theorem S7: Bayes selection and the information gap

For selectors measurable in $\mathcal{G}$, let $m_j = E[Y_j|\mathcal{G}]$. With finite K and measurable tie breaking,

$$\sup_{J \text{ } \mathcal{G}\text{-measurable}} E[Y_J] = E[\max_{1 \leq j \leq K} m_j] \leq E[\max_{1 \leq j \leq K} Y_j]. \quad (\text{S7})$$

Proof. Conditional on $\mathcal{G}$, any (possibly randomized) selector has mean $\sum_j P(J = j|\mathcal{G}) m_j$, at most the largest m_j ; the argmax attains it. For the second inequality, each $m_j \leq E[\max_l Y_l|\mathcal{G}]$; maximize and integrate. $\square$

An oracle pass-at- K gap is a ceiling, not a demonstrated recoverable gain. A null result for two fitted selectors means those fitted procedures did not demonstrate improvement with the available data. It does **not** prove $E \max_j m_j$ equals the baseline, prove all learning methods on the same features fail, or establish irreducible insufficiency of that feature map.

A sequential resampler can generate candidates one at a time, stop at a public signal, and return the current candidate; Theorem S1 applies to its own full resampling trajectories. If it can return any earlier stored candidate, define both a stopping time τ and an $\mathcal{F}_\tau^O$ -measurable terminal selector $J \leq \tau$; the path functional is $Y_J - \lambda C_\tau$. Identical coupling proves unweighted replay under a matching bank-generation mechanism. If it generates all K candidates before selecting, it incurs all K generation costs. Retrospectively finding an early successful candidate does not entitle a selector that inspected later candidates to the earlier cost.

To attribute performance to feedback content rather than selection or extra samples, independently compare frozen policies under matched information and total cost. A resampling selector’s success establishes no causal feedback benefit; a feedback policy’s failure against one resampler is evidence about that tested comparison, not a theorem that no future feedback policy can help.

6. What finite-sample guarantees and ESS bounds do—and do not—say

Theorem S8: honest finite-class unweighted evaluation

Condition on a training sample that fixes m candidate stopping/selection rules. Suppose each independent root yields a pre-specified averaged score for each rule in $[a, b]$. Let n be the number of independent identically sampled root clusters. With probability at least $1 - \alpha$, simultaneously,

$$|\widehat{V}_j - V_j| \leq (b - a) \sqrt{\frac{\log(2m/\alpha)}{2n}}, \quad 1 \leq j \leq m. \quad (\text{S8})$$

Proof. Each frozen rule's cluster scores are bounded independent draws. Hoeffding's inequality controls its two tails, and a union bound over m rules proves (S8). Within-cluster averaging preserves the range regardless of dependence within the root. $\square$

For paired rule differences in $[-B, B]$, substitute range $2B$, or use a separately justified variance-sensitive bound. Such worst-case intervals can be too broad to resolve a small effect. This establishes the limitations of **that bound**, not impossibility of every finite-sample method. As an explicit counterexample, take a paired difference $D_i \in \{-1, +1\}$ and suppose all n independent observations are $+1$. Under the null $ED_i \leq 0$, the success probability is at most one half, so the event has probability at most 2^{-n} . The test rejecting only on that event is valid and can certify a positive mean at $n=176$ (indeed much smaller), although a conservative multirule worst-case radius could be uninformative for the particular project gap. This example is not a claim that the project's observed differences have that distribution or evidence.

A finite-sample paired bootstrap interval is not automatically a nonasymptotic certificate. Its validity needs its own cluster-sampling, regularity, and estimator conditions. Evaluation of an independently frozen rule is a regular mean problem; estimating the **optimal** value after searching unrestricted rules can be nonregular near tied action values. Binary success probabilities near zero or one alone do not imply a tied action contrast or identify the mass of an exceptional law.

Proposition S9: overlap bounds are lower bounds on an idealized ESS

Suppose nonnegative stopped weights satisfy $E W = 1$ and $W \leq M$. Then $EW^2 \leq M$, so the population proxy

$$n_{\text{eff}}^{\text{pop}} = n \frac{(EW)^2}{EW^2} \quad \text{satisfies} \quad n_{\text{eff}}^{\text{pop}} \geq n/M. \quad (\text{S9})$$

Proof. $W^2 \leq MW$ pointwise; take expectations and rearrange. $\square$

For eight equally randomized actions and two deterministic overrides, the **worst-case** path weight is 64, so $176/64=2.75$ is a lower bound under the proxy, not the actual number of effective clusters. For example, a target only slightly different from the logging selector at both decisions has maximum bound 64 but second moment close to one, hence ESS close to 176. The empirical ratio $(\sum_i W_i)^2 / \sum_i W_i^2$ is random and is not guaranteed by the population bound to exceed n/M in every realized sample. Cluster averaging can alter the appropriate weights/variance calculation. Actual precision depends on scores, outcomes, cluster dependence, and covariance—not just the weight proxy.

Likewise a sufficient planning expression such as $n \geq 8M \log(4m/\alpha)/gap^2$ is not an information-theoretic necessary sample size unless accompanied by a matching lower-bound theorem for the exact utility and target. Clipping a weighted utility has sign-definite bias only under the stated utility sign restriction. A raw utility $Y - \lambda C$ can be negative; with $Y \in [0, 1]$ and $C \in [0, C_{\max}]$ its range length is $1 + \lambda C_{\max}$, not one unless explicitly normalized.

Proposition S10: hidden carried state does not automatically defeat randomized policy evaluation

Let L_k be unrecorded internal generator/receiver state. Suppose logging actions are genuinely randomized with known $e_k(a|H_k)$ independent of L_k conditional on the full recorded history, the target actions use $\pi_k(a|H_k)$, and both executions share the same initial latent-state distribution and state-transition kernel $K(dH_{k+1}, dL_{k+1} | H_k, L_k, A_k)$. Assume support. Then the joint latent-and-observed trajectory likelihood ratio is the observable product $\prod_k \pi_k/e_k$, and the same ratio identifies the marginal observed trajectory law after integrating out L .

Proof. Factor the augmented trajectory densities. The common initial law and every common latent/environment transition cancel, leaving only the action-probability ratios. This remaining product is measurable in the observed trajectory. Therefore, when marginalizing the latent path, it can be taken outside the integral and remains the marginal density ratio. $\square$

Thus the blanket claim that any carried cache, scratchpad, or action-dependent hidden memory necessarily prevents cancellation is too strong. Hidden memory creates real dangers when it affects assignment beyond logged propensities, changes mechanism across collection/deployment, is omitted from a history needed to reconstruct assignment, causes cross-root interference, or prevents faithful branch restoration. It can also obstruct a reduced-state transition model. Stateless generation is a useful sufficient engineering safeguard; byte-identical output under a stochastic generator is neither necessary for this theorem nor a general test of distributional equality.

7. Decision for Q11 and the next bounded experiment

Q11 resolution: the primary methodological claim should be **identification and honest evaluation of supported language-intervention regimes**, with horizon-aware stopping/selection as a concrete application. The current evidence does not support a headline that a new causal critic improves decisions over correctly history-adjusted sequential regression. Neither does the selection pilot yet establish a learned-selection improvement. This is a scope decision, not a declaration of empirical success or novelty priority.

The narrower immediately testable target is the out-of-task value of a frozen, public-information stopping/selection policy against a strong matched-cost baseline. If full rollouts match its continuation, evaluate by paired unweighted root means. If it changes feedback content, generation, or receiver routing, use a supported action-value design or fresh paired execution; do not label that comparison $w=1$ merely because STOP grades are observed.

The original goal of learning next-prompt values remains legitimate as a separate experiment: randomize supported candidate prompts from restored prefixes and evaluate specified continuations. It should not borrow evidence from independent-sample selection. A scope-limited negative result is an acceptable outcome. A claim that a whole feature family, all feedback policies, or all certificates are impossible requires substantially stronger analysis than the current pilot and worst-case inequalities.

8. Executable verification and remaining boundaries

[check_stopping_identities.py](#) enumerates an explicitly stated finite law and checks matching-continuation replay, changed-continuation weighting, oracle/public information separation, censoring correction, and sharp counterexamples to unsupported extrapolations. [test_stopping_theory.py](#) supplies regression checks. These verify algebra in the finite model; they do not establish real-log exchangeability, complete prefix preservation, feature sufficiency, or LLM efficacy.

No new neural convergence rate, universal confidence band, optimal generative prompt theorem, or empirical result is claimed here. The independent experimental audit must establish whether concrete logs satisfy S1–S5 before any real-data replay value receives the interpretation in (S1).

Corrected E0 simulation study, 2026-09-20

Executed: 17 cells × 80 replications = 1,360 replications, local CPU only. This is a corrected diagnostic grid, not the planned 1,000-replication-per-cell final precision study. Every replication retains its seed, fitted-Q results, exact actual-template-mixture truth, rule values and offsets. Each run includes a matching source snapshot. Original results are preserved; the Sept 19 interpretation is superseded.

What changed

The first-action inverse probability factor now restores the latent distribution within the initial state. Conditional raw regression is evaluated both against its own logging-continuation association and against the distinct uniform-continuation target. Template heterogeneity is graded against the replicate-specific mixture kernel, rather than the zero-offset kernel. The previously named state-only optimum is a heuristic reference; its difference from a learned rule is not optimal-policy regret. The driver now executes exactly the requested number of distinct seeds, independent of worker count, and saves per-replication records.

Fitted-Q policy-value results

Cell	Replications	DR bias	Bias MCSE	95% coverage	Coverage MCSE
FAIL_coarsening	80	-0.0029	0.0016	0.9625	0.0212
FAIL_latent	80	+0.0020	0.0018	0.9250	0.0294
base	80	-0.0020	0.0017	0.9625	0.0212
kappa0	80	-0.0002	0.0015	0.9625	0.0212
FAIL_mislabel	80	+0.0002	0.0017	0.9125	0.0316
FAIL_positivity	80	-0.0158	0.0086	0.8125	0.0436
T2	80	-0.0040	0.0018	0.9500	0.0244
T5	80	-0.0021	0.0018	0.9500	0.0244
floor0.02	80	+0.0014	0.0023	0.9000	0.0335
floor0.05	80	-0.0015	0.0016	0.9500	0.0244
half_effect	80	-0.0004	0.0018	0.9375	0.0271
kappa0.5	80	-0.0030	0.0017	0.9375	0.0271
kappa2	80	-0.0005	0.0018	0.9625	0.0212
latent_track	80	+0.0004	0.0015	0.9625	0.0212
n590	80	+0.0020	0.0010	0.9625	0.0212
runs10	80	+0.0021	0.0021	0.9000	0.0335
runs160	80	-0.0011	0.0014	0.9625	0.0212

Known logging probabilities are used throughout; the outcome regressions are actually fitted and cross-fitted by root task. Persistent latents make the compressed state Q potentially misspecified. In latent-dependent assignment cells, recorded B contains the true latent-dependent assignment probability: this is not identification from unknown confounding. No fitted behavior-model claim is made.

The weak-overlap cell still has a positive .02 probability floor, so it demonstrates practical finite-sample failure, not nonidentification from zero support. Its coverage is .8125 (MCSE .0436), with bias $-.0158$ (see table); Wald intervals cannot be assumed reliable there. The template-mixture cell now has .9625 coverage (MCSE .0212), using its actual law. This removes the earlier invalid interpretation that template variation intrinsically breaks a declared class-mixture intervention. Eighty replications remain too imprecise to establish nominal coverage at high precision. No multiplicity-selected winner is claimed.

Reproduce and extend

Run

```
uv run python experiments/e0/run_grid.py \  
  --reps 80 \  
  --workers 4 \  
  --seed-base 20260920 \  
  --out results/new_corrected_grid
```

. Seeds are tied to the stable cell registry, so the two executed batches and a combined invocation share seeds. Numerical source snapshots are under the two dated runs in `results/e0_corrected_20260920/`; timing/platform fields can differ.

Before using these as a final manuscript coverage study, freeze a desired Monte Carlo precision, increase replications accordingly, and include a correct full-history sequential-regression baseline for the same continuation target. A separate observational-data claim requires actually fitted propensity nuisances and their assumptions. The synthetic study does not establish real prompt efficacy.

Corrected fixed-bank selection pilot, 2026-09-20

Executed on existing data: zero model calls and zero GPU hours. All 4,488 initial candidates are retained, regardless of later routing: 561 tasks $\times$ two receivers $\times$ four draws. A frozen seeded split uses 336 task roots for fitting and 225 for evaluation, shared across receivers. This reuses a previously analysed corpus, including its former confirmatory labels, and is explicitly exploratory.

Each policy receives the same bank of four completed initial answers and public validation results. All four generation calls are charged to every selector. The visible comparator chooses uniformly among visible-passing candidates, or uniformly among all four if none passes. This equals averaging the first-visible-pass rule over random bank order. Learned policies maximize their fitted score with uniform tie-breaking. The endpoint is recorded **full-test success**, not independently rescored hidden-only performance.

Receiver	Selector	Visible success	Learned success	Difference (percentage points)	Exploratory task 95% interval (pp)
qwen2.5-3b-instruct	gbm	0.6674	0.6700	+0.26	[-1.92, +2.44]
qwen2.5-3b-instruct	logit	0.6674	0.6667	-0.07	[-2.48, +2.34]
qwen2.5-7b-instruct	gbm	0.7093	0.7122	+0.30	[-1.59, +2.19]
qwen2.5-7b-instruct	logit	0.7093	0.7378	+2.85	[+1.19, +4.52]

The 7B logistic selector is a promising signal: +2.85 percentage points for this frozen-bank task split. This is not a confirmed effect after adaptive project-level selection or multiplicity correction. It does not establish a cheaper policy, multi-round feedback benefit, causal next-prompt value, or a learned dynamic regime. The other three fits provide little evidence of improvement. We retain all four comparisons rather than reporting only the positive one.

Integrity and limits

An independent internal agent reconstructed all 900 per-task rows from the raw source and checked source/script hashes, split membership, labels, exact tie expectations, generation-token totals, paired means, standard errors and intervals; they matched. No direct hidden-label feature leakage was found. This is an internal audit, not external replication or proof of data validity.

Intervals condition on the frozen training data and assume independent task roots. Benchmark families remain unresolved, all project-level analyses have already inspected the corpus, and full-test labels can overlap prompt-visible material. Standardization fits training data only. Generation prompt/completion costs match by construction; checker and selector CPU time were not measured. There is no comparison at the lower adaptive-BoN stopping cost.

Inherited failure-class features name `assert` while the source emits `assertion`; that intended one-hot is therefore absent. The public visible-pass and fraction-failed features still record failure. Preserve this run as executed; fix the schema prospectively before a new frozen experiment. Do not silently refit around the positive result.

An initial run aborted before fitting because a strict validator expected a Boolean but the recorded outcome is binary integer. The corrected run accepts only Boolean or integer 0/1 outcomes; the failure record is retained. Source logs were not edited.

Reproduction

```
uv run python scripts/recheck_selection.py \  
  --episodes /path/to/DTR-AgentEvals/results/code_routing/log/episodes.jsonl \  
  --out results/new_selection_recheck
```

Use the source SHA256 in `results/selection_corrected_20260920_v2/manifest.json`. The complete manifest, task split, hyperparameters, predicted scores, per-task outcomes and summaries are retained. The source corpus is not bundled into this repository.

Next falsifiable experiment

Freeze logistic selection, visible selection and adaptive-BoN baselines on a new root/family-separated task set. First compare fixed banks at identical generation budgets; separately compare a prespecified sequential policy at a matched expected-call/token budget, measuring selector/checker costs too. Audit visible versus hidden test separation and invalid-test rates before generation. Preserve failures. Do not relabel this candidate-selection follow-up as evidence for prompt treatment effects; that requires its own randomized feedback experiment.